NAME

compare − longest common subsequence of two files

SYNOPSIS

compare [*options*] *file1 file2*

DESCRIPTION

compare computes the Longest Common Subsequence (LCS) of *file1* and *file2*. By default it prints their differences as an edit script; with **−c** it prints the common (matching) records instead, and with **−l** it prints only the LCS length.

Each difference is introduced by a header line:

<<<<< n *file1* [*begin*:*end*] <<<<<
(records present only in *file1*)
>>>>> n *file2* [*begin*:*end*] >>>>>
(records present only in *file2*)

where *n* is the one-based difference number and *[begin:end]* the half-open range of records involved. A final line reports the LCS length, and when compiled with duration reporting enabled, a timing line such as *RickSpace: 28 operations in 9 µs @3.111 MHz* follows.

The two files are compared at one of three granularities:

record each input line is one record (the default);

text (**−t**) the whole file is one sequence of Unicode code points, so multi-byte characters are never split;

word (**−w**) each whitespace-separated word is one record.

Three LCS algorithms are available through **−a**:

d Hunt−Szymanski (1977), the default: a threshold array over the match lists, O((r+n) log n) time.

s RickSpace (Rick 2000): forward/backward contours plus Hirschberg recursion, O(ns) space.

t RickTime (Rick 1995): a single diagonal sweep, O(ns + d) space.

OPTIONS

−a *d*|*s*|*t*
Select the LCS algorithm: **d** (Hunt−Szymanski, the default), **s** (RickSpace), or **t** (RickTime). Accepts **−ad** or **−a d**.

−b Ignore whitespace: collapse each run of whitespace to a single space before comparing, and expand the reported spans back to the original positions.

−c Print the correspondence (the common, matching records) instead of the differences.

−e Include the host environment (locale, processor, and memory) in the output.

−f Treat the input files as FASTA and strip header lines; implies **−t**.

−h Print a usage synopsis and exit.

−i Perform a case-insensitive comparison.

−j *join* Coalesce nearby differences that lie within *join* records of each other.

−l Report only the LCS length; skip reconstructing the differences.

−p *prefix*
Print *prefix* lines of context before each difference.

−q Treat *file1* and *file2* as literal strings rather than filenames; implies **−t**.

−s *suffix*
Print *suffix* lines of context after each difference.

−t Compare the whole file as a sequence of Unicode code points (the text path) rather than line-by-line records.

−V Print the version and exit.

−v Enable verbose output: print the LCS length, the start and finish wall-clock times, and a timing summary; by default only the differences are shown.

−w Split each line into whitespace-separated words and compare word by word (record mode).

EXIT STATUS

0 Success (including **−h** and **−V**).

1 Error: a command-line parse failure, a missing file, or an algorithm failure.

ENVIRONMENT

The locale (in particular **LANG**, **LC_ALL**, and **LC_NUMERIC**) controls the thousands separator used in the reported counts and the locale name shown by **−e**. On a system whose locale defines digit grouping, counts are printed with comma separators (for example **LCS Length = 4,591,960**); otherwise they are printed ungrouped.

EXAMPLES

Compare two files line by line and show the differences:

```
compare old.txt new.txt
```

Compare as text (code points), case-insensitively, showing only the length:

```
compare −i −l −t a.fa b.fa
```

Compare two FASTA genomes with the linear-space algorithm and show matches:

```
compare −as −c −f ecoli.seq ecoli_100subs.seq
```

Compare two literal strings rather than files:

```
compare −q "the quick sly fox jumped" "over the lazy brown dog"
```

SEE ALSO

delta(1), **diff**(1), **cmp**(1)

The LCS algorithms and their performance bounds are described in *LCSRickSpace.md* and *LCSRick-Time.md* in the project documentation.